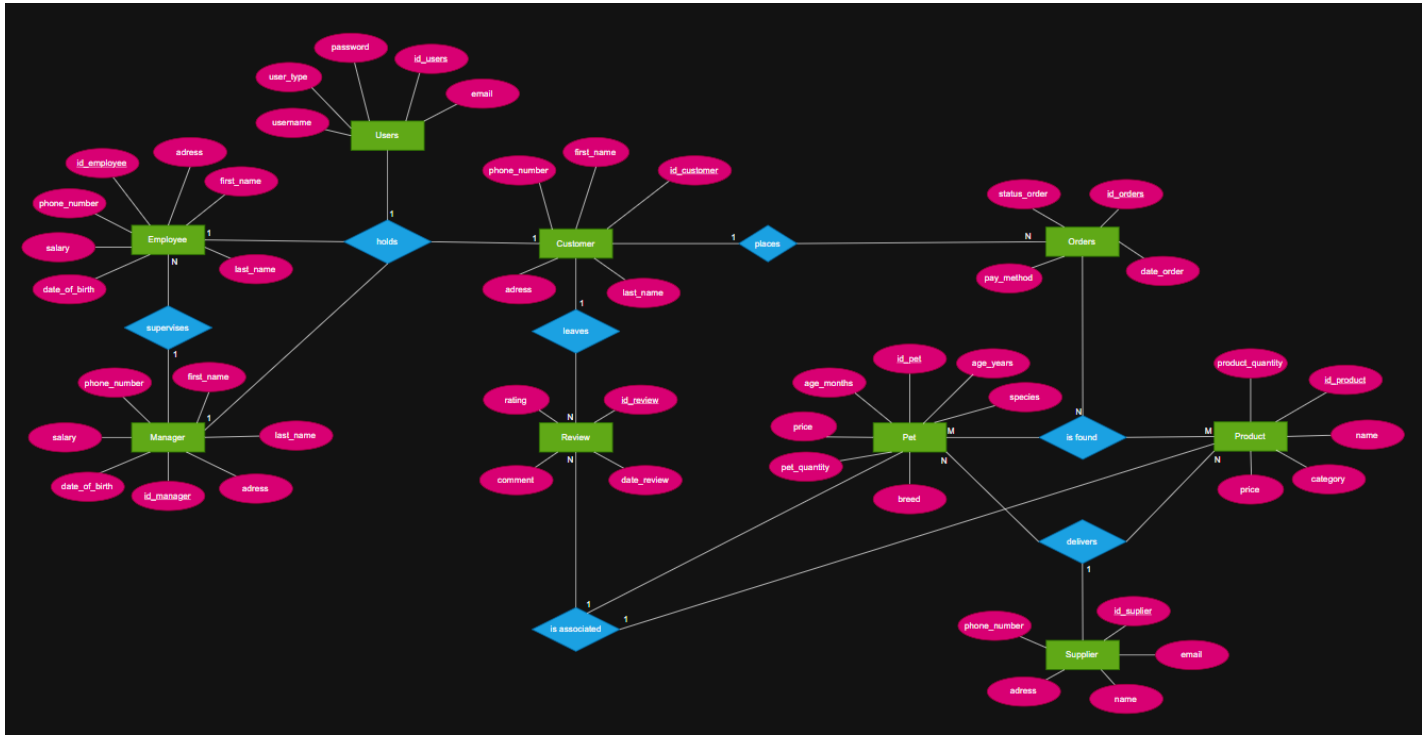
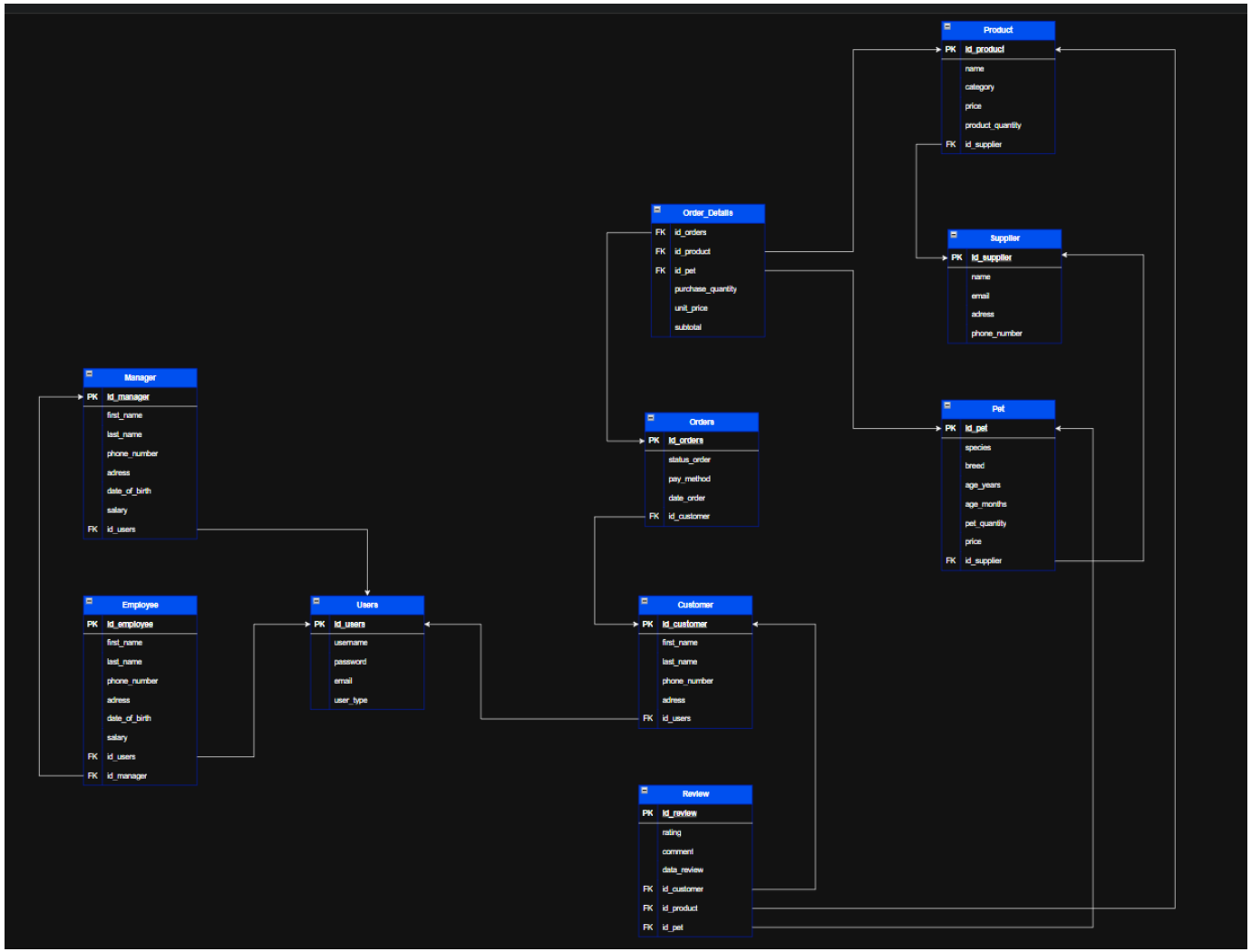
Proiect Baze de date

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Specializarea:Calculatoare(Română)
Grupa:CR3.1A
Temă proiect:PetShop



Modelul entitate-legătură



Modelul relațional

Entități

- users

-atribute

- id_user
- username
- password
- email
- user_type

- customer

-atribute:

- id_customer (primary key)
- first_name
- last_name
- phone_number
- adress
- id_users (foreign key)

- manager

-atribute

- id_manager (primary key)
- first_name
- last_name
- phone_number
- adress
- date_of_birth
- salary

- id_users (foreign key)

- employee

-atribute

- id_employee (primary key)
- first_name
- last_name
- phone_number
- adress
- date_of_birth
- salary
- id_users (foreign key)
- id_manager (foreign key)

- supplier

-atribute

- id_supplier(primary key)
- name
- email
- adress
- phone_number

- pet

-atribute

- id_pet(primary key)
- species
- breed
- age_years

- age_months
- price
- pet_quantity
- id_supplier(foreign key)

- product

-atribute

- id_product(primary key)
- name
- category
- price
- product_quantity
- id_supplier(foreign key)

- review

-atribute

- id_review (primary key)
- rating
- comment
- date_review
- id_customer (foreign key)
- id_pet (foreign key)
- id_product (foreign key)

- orders

-atribute

- id_orders (primary key)
- status_order

- pay_method
- date_order
- id_customer (foreign key)

- order_details(tabel de asociere)

-atribute

- id_orders (foreign key)
- id_pet (foreign key)
- id_product (foreign key)
- purchase_quantity
- unit_price
- subtotal

Crearea tabelelor

```
create table users(
id_users serial primary key,
username varchar(50) not null,
password varchar(50) not null,
email varchar(50) not null unique,
user_type varchar(50) check(user_type in('customer','employee','manager'))
);

create table manager(
id_manager serial primary key,
first_name varchar(50) not null,
last_name varchar(50) not null,
phone_number int not null unique,
```

```
adress varchar(50),
date_of_birth date,
salary int,
id_users int,
foreign key(id_users) references users(id_users) on delete cascade on update cascade
);

create table employee(
id_employee serial primary key,
first_name varchar(50) not null,
last_name varchar(50) not null,
phone_number int not null unique,
adress varchar(50),
date_of_birth date,
salary int,
id_users int,
id_manager int,
foreign key(id_users) references users(id_users) on delete cascade on update cascade,
foreign key(id_manager) references manager(id_manager) on delete cascade on update cascade
);

create table customer(
id_customer serial primary key,
first_name varchar(50) not null,
last_name varchar(50) not null,
```

```
phone_number int not null unique,
adress varchar(50),
id_users int,
foreign key(id_users) references users(id_users) on delete cascade on update cascade
);

create table supplier(
id_supplier int primary key,
name varchar(50) not null,
email varchar(50) not null unique,
adress varchar(50),
phone_number int not null  unique
);

create table pet(
id_pet int primary key,
species varchar(50) not null,
breed varchar(50) not null,
age_years int check(age_years>=0),
age_months int check(age_months>=0 and age_months<12),
price int not null,
pet_quantity int,
id_supplier int,
foreign key(id_supplier) references supplier(id_supplier) on delete cascade on update cascade
);
```

```
create table product(
id_product int primary key,
name varchar(50) not null,
category varchar(50) not null,
price int not null,
product_quantity int,
id_supplier int,
foreign key(id_supplier) references supplier(id_supplier) on delete cascade on update cascade
);

create table review(
id_review int primary key,
rating int check(rating>=0 and rating<=5),
comment varchar(50),
date_review date,
id_customer int,
id_pet int,
id_product int,
check((id_pet is not null) or (id_product is not null)),
foreign key(id_customer) references customer(id_customer) on delete cascade on update cascade,
foreign key(id_pet) references pet(id_pet) on delete cascade on update cascade,
foreign key(id_product) references product(id_product) on delete cascade on update cascade
);
```

```
create table orders(
id_orders int primary key,
status_order varchar(50),
pay_method varchar(50),
date_order date,
id_customer int,
foreign key(id_customer) references customer(id_customer) on delete cascade on update cascade
);

create table order_details(
id_orders int,
id_pet int,
id_product int,
purchase_quantity int,
unit_price int,
subtotal int generated always as (purchase_quantity * unit_price) stored,
check((id_pet is not null) or (id_product is not null)),
foreign key(id_orders) references orders(id_orders) on delete cascade on update cascade,
foreign key(id_pet) references pet(id_pet) on delete cascade on update cascade,
foreign key(id_product) references product(id_product) on delete cascade on update cascade
);
```

Afișarea tabelelor

```
select * from users;
select * from customer;
select * from employee;
select * from manager;
select * from supplier;
select * from pet;
select * from product;
select * from review;
select * from orders;
select * from order_details;
```

Exemple de relații care nu respectă formele normale

Forma normală 1(FN1)

Exemplu

Employee

id_employee	first_name	last_name	phone_number	adress
1	marius	sandu	077237122	{Craiova,Strada Babadag,Nr.21}
2	camelia	marcu	073422100	{Brașov,Strada Iazul Verde,Nr.18}

Forma normală 2(FN2)

Exemplu

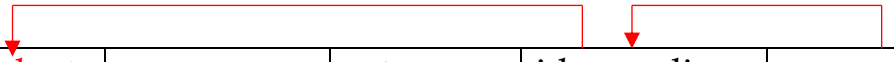
Pet

id_pet	id_customer	species	breed	phone_number
3	5	dog	Pitbull	0723100098
4	8	cat	Siamese	0725332134

Forma normală 3(FN3)

Exemplu

Product



id_product	name	category	id_supplier	name_supplier
10	dog food	pet food	2	Animalix
7	cat food	pet food	4	My Pet

Forma normală Boyce-Codd(BCNF)

Exemplu

Users

id_users	username	password	email	user_type →	role
6	radu23	pass24	radu_ion@gmail.com	employee	cashier
7	mihai12	pass27	mihai_david@gmail.com	manager	organizing

Forma normală 4(FN4)

Exemplu

Customer

id_customer	product	pet
8	aquarium	fish
9	dog food	dog

Inserări,modificări,ștergeri

- inserări în tabela users

```
insert into users(username, password, email, user_type) values
('andrei21','password1234','andrei_bitoleanu@gmail.com','customer'),
('sebi42','password2345','sebi_marcel@gmail.com','employee'),
('elena02','password1235','elena_popescu@gmail.com','customer'),
('cazimir41','password1222','cazimir_ionel@gmail.com','manager'),
('sorin89','password2223','sorin_tătaru@gmail.com','manager'),
('gigel54','password3345','gigel_enescu@gmail.com','customer'),
('ana19','password8745','ana_dumitrașcu@gmail.com','customer'),
('teodora65','password6754','teodora_anghel@gmail.com','employee'),
('john35','password1000','john_davis@gmail.com','employee'),
('anca27','password9324','anca_istrate@gmail.com','manager'),
('florin50','password3293','florin_andronie@gmail.com','manager'),
('eduard02','password2293','eduard_drăcea@gmail.com','employee'),
('filip62','password9000','filip_voiculescu@gmail.com','customer'),
('frank22','password3314','frank_johnson@gmail.com','customer'),
('mike77','password5633','mike_harold@gmail.com','employee'),
('jack11','password9721','jack_maurice@gmail.com','manager'),
('vladimir32','password9133','vladimir_putin@gmail.com','employee'),
('daniela45','password0032','daniela_georgescu@gmail.com','manager'),
('mirela20','password0011','mirela_zamfir@gmail.com','manager'),
('andrada76','password472','andrada_chivu@gmail.com','employee'),
('orlando81','password449','orlando_nedelea@gmail.com','customer'),
```

('georgel123','password168','georgel_lungu@gmail.com','customer');

- inserări în tabela customer

insert into customer(first_name, last_name, phone_number, adress, id_users) values
('Andrei','Bitoleanu', 773345643, 'Golumbu', 1),
('Elena','Popescu', 745234123, 'Craiova', 3),
('Gigel','Enescu', 724390784, 'Craiova', 6),
('Ana','Dumitraşcu', 776873454, 'Bucureşti', 7),
('Filip','Voiculescu', 773216734, 'Bucureşti', 13),
('Frank','Johnson', 771232100, 'Bucureşti', 14),
('Orlando','Nedelea', 700423210, 'Sibiu', 21),
('Georgel','Lungu', 711120755, 'Golumbu', 22);

- inserări în tabela manager

insert into manager(first_name, last_name, phone_number, adress, date_of_birth, salary, id_users) values
('Cazimir', 'Ionel', 723456789, 'Ilfov', '1980-05-15', 8000, 4),
('Sorin', 'Tătaru', 741234567, 'Craiova', '1975-10-10', 8500, 5),
('Anca', 'Istrate', 765678901 , 'Şimnic', '1982-08-22', 8200, 10),
('Florin', 'Andronie', 789345678, 'Şimnic', '1979-11-18', 8700, 11),
('Jack', 'Maurice', 700452832, 'Ilfov', '1982-07-05', 8900, 16),
('Daniela', 'Georgescu', 701234567, 'Melineşti', '1984-03-04', 8300, 18),
('Mirela', 'Zamfir', 791222734, 'Craiova', '1981-06-01', 8600, 19);

- inserări în tabela employee

insert into employee(first_name, last_name, phone_number, adress, date_of_birth, salary, id_users, id_manager)
values
('Sebi', 'Marcel', 743458309, 'Craiova', '1990-04-10', 4000, 2, 1),

('Teodora', 'Anghel', 723111234, 'Craiova', '1992-07-15', 4200, 8, 1),
('John', 'Davis', 734800645, 'Strehaia', '1991-03-20', 4100, 9, 2),
('Eduard', 'Drăcea', 789433222, 'Filiaşi', '1989-11-01', 4300, 12, 3),
('Mike', 'Harold', 723456643, 'Filiaşi', '1974-10-12', 4500, 15, 4),
('Vladimir', 'Putin', 700499643, 'Filiaşi', '1994-9-12', 4500, 17, 5),
('Andrada', 'Chivu', 748834567, 'Melineşti', '1986-08-24', 4400, 20, 3);

- inserări în tabela supplier

insert into supplier(id_supplier, name, email, adress, phone_number) values
(1, 'Pet Supplies', 'contact@petsupplies.com', 'Blvd Tudor Arghezi', 72223456),
(2, 'Happy Pets', 'contact@happypets.com', 'Cartierul 1Mai', 87657421),
(3, 'Animal World', 'info@animalworld.com', 'Blvd Ecaterina Teodorescu', 123256543),
(4, 'Animalix', 'info@animalix.com', 'Strada Ion Matei', 77775654),
(5, 'My Pet', 'service@mypet.com', 'Strada Trei Pistoale', 96355353),
(6, 'Maxi Pet', 'service@maxipet.com', 'Cartierul Dumbrava Nouă', 96355009),
(7, 'Pisi Cuţu', 'contact@pisicuţu.com', 'Strada Nicolae Bălcescu', 78832566);

- inserări în tabela pet

insert into pet(id_pet, species, breed, age_years, age_months, price, pet_quantity, id_supplier) values
(1, 'Dog', 'Golden Retriever', 2, 6, 800, 3, 1),
(2, 'Dog', 'Doberman', 3, 0, 500, 5, 2),
(3, 'Cat', 'Sphynx', 1, 2, 300, 2, 2),
(4, 'Snake', 'Python', 0, 8, 200, 10, 4),
(5, 'Fish', 'Clownfish', 1, 4, 100, 4, 6),
(6, 'Dog', 'German Shepherd', 4, 3, 700, 2, 2),
(7, 'Cat', 'Maine Coon', 2, 7, 600, 3, 5),
(8, 'Spider', 'Tarantula', 5, 0, 150, 6, 4),

(9, 'Cat', 'Siamese', 2, 1, 600, 7, 5),
(10, 'Fish', 'Goldenfish', 0, 3, 150, 12, 5),
(11, 'Bird', 'Parrot', 2, 5, 130, 5, 6),
(12, 'Bird', 'Canary', 1, 8, 100, 19, 6),
(13, 'Lizard', 'Iguana', 4, 0, 220, 3, 5),
(14, 'Dog', 'Beagle', 4, 2, 50, 10, 7),
(15, 'Lizard', 'Chameleon', 6, 5, 350, 4, 3),
(16, 'Dog', 'Bulldog', 0, 9, 260, 9, 3),
(17, 'Rabbit', 'Lionhead', 1, 2, 100, 10, 1);

 inserări în tabela product

insert into product(id_product, name, category, price, product_quantity, id_supplier) values
(1, 'Dog Food', 'Food', 20, 20, 1),
(2, 'Cat Food', 'Food', 30, 25, 1),
(3, 'Bird Seeds', 'Food', 15, 30, 1),
(4, 'Aquarium', 'Accessories', 100, 10, 4),
(5, 'Cat Cage', 'Accessories', 200, 13, 4),
(6, 'Leash', 'Accessories', 25, 30, 2),
(7, 'Dog Cage', 'Accessories', 175, 25, 3),
(8, 'Shampoo Dog', 'Grooming Products', 50, 17, 5),
(9, 'Flea and Tick Spray', 'Grooming Products', 60, 45, 6),
(10, 'Fish Food', 'Food', 30, 19, 5),
(11, 'Shampoo Cat', 'Grooming Products', 50, 20, 7),
(12, 'Dog Toys', 'Toys', 35, 15, 7),
(13, 'Cat Toys', 'Toys', 45, 20, 2),
(14, 'Bird Cage', 'Accesories', 150, 10, 6),

(15, 'Nail Clipper', 'Grooming Products', 50, 10, 4),
(16, 'Collar', 'Accessories', 12, 34, 4),
(17, 'Terrarium', 'Accessories', 135, 15, 3);

- inserări în tabela review

```
insert into review (id_review, rating, comment, date_review, id_customer, id_pet, id_product) values
(1, 5, 'My Golden Retriever is amazing!', '2023-01-01', 1, 1, NULL),
(2, 4, 'The dog food is great quality.', '2024-03-02', 2, NULL, 1),
(3, 3, 'The aquarium is decent but a bit expensive.', '2025-11-23', 2, NULL, 4),
(4, 5, 'My Maine Coon is so fluffy and adorable!', '2025-06-14', 4, 7, NULL),
(5, 2, 'The flea spray didn’t work as expected.', '2023-09-05', 6, NULL, 9),
(6, 4, 'The cat cage is very sturdy.', '2020-10-01', 3, NULL, 5),
(7, 5, 'The Clownfish looks amazing in the aquarium!', '2025-01-07', 1, 5, NULL),
(8, 2, 'The Tarantula looks scary.', '2024-02-03', 3, 8, NULL),
(9, 3, 'The shampoo is good.', '2025-03-03', 5, NULL, 11),
(10, 1,'Poor quality nail clipper,broke after one use', '2024-03-03', 8, NULL, 15);
```

- inserări în tabela orders

```
insert into orders(id_orders, status_order, pay_method, date_order, id_customer) values
(1, 'Completed', 'Credit Card', '2024-02-01', 1),
(2, 'Processing', 'Cash', '2024-07-05', 2),
(3, 'Completed', 'Debit Card', '2024-07-09', 2),
(4, 'Pending', 'Cash', '2025-01-04', 4),
(5, 'Returned', 'Credit Card', '2025-10-05', 5),
(6, 'Processing', 'Credit Card', '2024-03-06', 5),
(7, 'Pending', 'Debit Card', '2025-01-10', 3),
(8, 'Processing', 'Credit Card', '2023-01-09', 2),
```

(9, 'Completed', 'Cash', '2023-11-13', 1),
(10, 'Completed', 'Cash', '2024-09-23', 4);

- inserări în tabela order_details

```
insert into order_details(id_orders, id_pet, id_product, purchase_quantity, unit_price) values
(1, NULL, 1, 5, 20),
(2, 1, NULL, 1, 800),
(2, NULL, 7, 1, 175),
(3, NULL, 6, 1, 25),
(3, NULL, 8, 2, 50),
(8, 12, NULL, 2, 100),
(5, 4, NULL, 1, 200),
(5, 5, NULL, 1, 100),
(4, NULL, 12, 2, 35),
(4, NULL, 1, 2, 20),
(4, NULL, 13, 1, 45),
(10, 9, NULL, 1, 600),
(10, NULL, 2, 3, 30),
(10, NULL, 5, 1, 200);
```

- modificări

```
alter table manager
add constraint unique_id_user unique (id_users);

alter table employee
add constraint unique1_id_user unique (id_users);
```

```
alter table customer
add constraint unique2_id_user unique (id_users);

alter table supplier add contract boolean;

alter table supplier drop contract;

update pet set price=250
where breed='Iguana'

update product set product_quantity = 14
where price between 90 and 160
```

- modificări care utilizează subinterogări

```
update product set price = price * 1.1
where id_supplier = (
   select id_supplier
   from supplier
   where name = 'Happy Pets'
);

update employee set salary = salary + 500
where id_manager = (
   select id_manager
   from manager
   where last_name = 'Tătaru'
```

```
);

update manager set date_of_birth = '1989-07-05'
where id_users = (
    select id_users
        from users
    where username like 'da%'
);
```

- ștergeri

```
delete from orders
where date_order = '2023-11-13';

delete from pet
where price between 30 and 60;
```

- ștergeri care utilizează subinterogări

```
delete from review
where id_customer in (
    select id_customer
        from customer
    where adress like 'Sibiu'
);

delete from product
where id_product in(
   select id_product
```

```
  from review
   where rating < 3
);
```

Interogări

a.interogari cu operatii de jonctiune (inner join+outer join)

- Să se afişeze emailurile angajaţilor împreună cu numele managerilor care îi supervizează.

```
select e.first_name as employee_first_name, e.last_name as employee_last_name, u.email as employee_email,
        m.first_name as manager_first_name, m.last_name as manager_last_name
from employee e
inner join users u on e.id_users = u.id_users
inner join manager m on e.id_manager = m.id_manager;
```

- Să se afişeze numele clienţilor care au plasat comenzi şi statusul comenzilor.

```
select c.first_name as customer_first_name, c.last_name as customer_last_name, o.status_order
from customer c
inner join orders o on c.id_customer = o.id_customer;
```

- Să se afişeze numele clientilor care au comandat produse cu preţul cuprins între 30 şi 70.

```
select c.first_name as customer_first_name, c.last_name as customer_last_name, p.name as product_name,
p.price
from customer c
inner join orders o on c.id_customer = o.id_customer
```

```
inner join order_details od on o.id_orders = od.id_orders
inner join product p on od.id_product = p.id_product
where p.price between 30 and 70;
```

- Să se afişeze toţi managerii,chiar dacă nu au angajaţi de supervizat

```
select m.first_name as manager_first_name, m.last_name as manager_last_name,
        e.first_name as employee_first_name, e.last_name as employee_last_name
from manager m
left outer join employee e on m.id_manager = e.id_manager;
```

- Să se afişeze toţi clienţii care au lăsat sau nu un feedback

```
select first_name,last_name,comment
from customer
left outer join review on customer.id_customer = review.id_customer;
```

- Să se afişeze clienţii din Bucureşti care au plasat sau nu comenzi

```
select c.first_name, c.last_name, c.adress, o.id_orders, o.status_order, o.date_order
from customer c
left outer join orders o on c.id_customer = o.id_customer
where c.adress like 'Bucureşti%';
```

b. interogari care sa contina functii de agregare si Group by

- Să se afişeze numărul total de animale al fiecărei specii

```
select species, count(*) as total_animals
from pet
group by species;
```

- Să se afişeze preţul mediu al produselor din fiecare categorie

```
select category, avg(price) as average_price
from product
group by category;
```

- Să se afişeze suma salariilor angajaţiilor,salariul maxim şi salariul minim

```
select sum(salary) as sum_salary, max(salary) as max_salary, min(salary) as min_salary
from employee;
```

- Să se afişeze animalele care s-au vândut,numărul lor cât şi suma totală

```
select p.species, count(od.purchase_quantity) as total_sold, sum(subtotal) as total_value
from pet p
inner join order_details od on p.id_pet = od.id_pet
group by p.species;
```

c.exemple de subinterogari (IN , ANY, Exists, etc)

- Subinterogari corelate în care intervin cel puţin 3 tabele

Să se selecteze produsele care au recenzie de 3 stele

```
select p.name, p.category,re.rating
from product p
inner join review re on p.id_product=re.id_product
where exists (
   select *
   from review r
   where r.id_product = p.id_product and r.rating = 3
);
```

Să se afişeze furnizorii care au livrat produse comandate de clientul cu id-ul 2

```
select s.name
from supplier s
where exists (
   select *
   from product p
   join order_details od on p.id_product = od.id_product
   join orders o on od.id_orders = o.id_orders
   where p.id_supplier = s.id_supplier and o.id_customer = 2
);
```

- Să se afişeze produsele care au preţul mai mare decât cel puţin preţul unui câine şi sunt furnizate de furnizorul 'Animalix'

```
select p.name, p.price
from product p
where p.price > any (
   select pe.price
   from pet pe
   where pe.species = 'Dog'
)
and exists (
   select *
   from supplier s
   where s.id_supplier = p.id_supplier
   and s.name = 'Animalix'
);
```

- Subinterogări necorelate în care intervin cel puţin 3 tabele
- Să se afişeze numele şi prenumele clienţilor care au comandat produse din categoria 'Food'

```
select c.first_name, c.last_name
from customer c
where c.id_customer in (
   select o.id_customer
   from orders o
```

```
    join order_details od on o.id_orders = od.id_orders
    join product p on od.id_product = p.id_product
    where p.category = 'Food'
);
```

- Să se afişeze numele şi prenumele clienţilor care au lăsat recenzii la animale cu preţul mai mare de 600

```
select c.first_name, c.last_name
from customer c
where exists (
    select *
    from review r
    where r.id_customer = c.id_customer
    and exists (
        select *
        from pet p
        where p.id_pet = r.id_pet
        and p.price > 600
    )
);
```

- Să se afişeze id-ul comenzii şi data comenzii pentru toate comenzile plasate de clienţii care locuiesc în Bucureşti şi care conţin produse din categoria 'Food'.

```
select o.id_orders, o.date_order
```

```
from orders o
where o.id_customer = any (
   select c.id_customer
   from customer c
   where c.adress like 'Bucureşti'
)
and o.id_orders in (
   select od.id_orders
   from order_details od
   join product p on od.id_product = p.id_product
   where p.category = 'Food'
);
```

d.

- utilizarea a cel puţin două funcţii pe şiruri de caractere

```
select
   concat(c.first_name, ' ', c.last_name) as full_name,
   upper(c.adress) as upper_adress,
   length(c.first_name) as first_name_length
from
   customer c
where
   c.first_name like 'A%';
```

- utilizarea a cel puţin două funcţii pe date calendaristice

```
select
   o.id_orders,
   o.date_order,
   current_date - o.date_order as days_since_order,
   extract(year from o.date_order) as order_year
from
   orders o
where
   o.date_order between '2024-01-01' and '2024-12-31';
```

- utilizarea a cel puțin unei expresii case

```
select
   p.name,
   p.price,
   case
      when p.price > 100 then 'Expensive'
      when p.price between 50 and 100 then 'Moderate'
      else 'Cheap'
   end as price_category
from
   product p;

select
   e.first_name,
   e.last_name,
```

```
    e.salary,
    case
        when e.salary >= 4500 then 'high salary'
        when e.salary between 4300 and 4400 then 'medium salary'
        else 'low salary'
    end as salary_type,
    case
        when extract(month from e.date_of_birth) = 3 then 'march'
        when extract(month from e.date_of_birth) = 4 then 'april'
        else 'unknown'
    end as birth_month
from
    employee e;
```

e.vederi

```
create view total_order_amount as
select o.id_orders,c.first_name,c.last_name,sum(od.subtotal) as total_amount
from  orders o
inner join customer c on o.id_customer = c.id_customer
inner join order_details od on o.id_orders = od.id_orders
group by o.id_orders, c.first_name, c.last_name;
```

```
create view pet_details as
select p.id_pet,p.species,p.breed,p.price,s.name as supplier_name,s.phone_number as supplier_contact
from pet p
inner join supplier s on p.id_supplier = s.id_supplier;

create view user_details as
select
   u.id_users,
   u.username,
   u.email,
   case
      when u.user_type = 'customer' then (select c.first_name || ' ' || c.last_name from customer c where
c.id_users = u.id_users)
      when u.user_type = 'employee' then (select e.first_name || ' ' || e.last_name from employee e where
e.id_users = u.id_users)
      when u.user_type = 'manager' then (select m.first_name || ' ' || m.last_name from manager m where
m.id_users = u.id_users)
   end as full_name,
       u.user_type
from  users u;
```

- exemplu de operație LMD permisă pe vedere

```
create view v_pet_details as
select id_pet, species, breed, price
from pet;
```

```
update v_pet_details
set price = 850
where id_pet = 1;
```

- exemplu de operaţie LMD nepermisă pe vedere

```
create view v_pet_sum as
select species, avg(price) as avg_price
from pet;

update v_pet_sum
set avg_price = 500
where species = 'Dog';
```

Crearea unui index care să optimizeze o interogare cu 2 criterii de cautare

```
select *
from employee
where last_name = 'Putin'
and first_name = 'Vladimir';

create idx_employee_lastname_firstname
on employee (last_name, first_name);
```